Unit: Unit 3: Functions
Topic: Composite Functions

Name: ________________
Date: ____________

Homework (Habanero)

Q1. If $f(x) = x^2$ and $g(x) = 2x + 1$, what is $f(g(x))$?
(A) $4x^2 + 4x + 1$
(B) $4x^2 + 2$
(C) $2x^2 + 2x + 1$
(D) $x^2 + 2x + 1$
(E) $2x^2 + 1$

Q2. Given $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x+1}$, what is the domain of $f(g(x))$?
(A) $x \neq -1, x \neq 0$
(B) $x \neq 0$
(C) $x \neq -1$
(D) $x > 0$
(E) $x < 0$

Q3. A patient's dosage of medicine is determined by the function $f(x) = 2x$, where x is the patient's weight. If $g(x)$ represents the patient's weight after a 10
(A) $1.8x$
(B) $2x$
(C) $2.2x$
(D) $1.9x$
(E) $2.1x$

Q4. The growth rate of a cell culture is modeled by $f(x) = 3^x$, where x is time in hours. If $g(x) = x - 2$, what does $f(g(x))$ represent?
(A) The growth rate after a 2-hour delay
(B) The growth rate 2 hours ago
(C) The growth rate in 2 hours
(D) The initial growth rate
(E) The final growth rate

Q5. A biologist is studying the effect of a new drug on bacteria growth. The function $f(x) = \frac{1}{x}$ models the growth rate, and $g(x) = x + 2$ models the change in environment. What is $f(g(x))$?
(A) $\frac{1}{x+2}$
(B) $\frac{1}{x}$
(C) $\frac{1}{x-2}$
(D) $\frac{x+2}{x}$
(E) $\frac{x-2}{x}$

Q6. If $f(x) = \sqrt{x}$ and $g(x) = x^2 - 4$, what is the domain of $f(g(x))$?
(A) $x \geq 2, x \leq -2$
(B) $x > 2$
(C) $x \geq 0$
(D) $x \geq -2$
(E) $x < 0$

Q7. A medical researcher is analyzing the relationship between the dosage of a new vaccine and its effectiveness. The function $f(x) = 0.8x$ represents the dosage, and $g(x) = x + 10$ represents the patient's immune response. What is $f(g(x))$?
(A) $0.8x + 8$
(B) $0.8x + 10$
(C) $0.8x - 8$
(D) $0.8x - 10$
(E) $0.8x$

Q8. The function $f(x) = 2x$ models the rate at which a patient recovers from a disease, and $g(x) = x - 5$ models the time elapsed since treatment began. What is $f(g(x))$?
(A) $2x - 10$
(B) $2x + 5$
(C) $2x - 5$
(D) $2x + 10$
(E) $2x$

Open Ended Questions (FRQ)

Q9. A pharmaceutical company is developing a new medication that affects the body's production of a certain enzyme. The function $f(x) = 0.5x$ models the amount of enzyme produced, and $g(x) = 2x + 1$ models the dosage of the medication. Find $f(g(x))$ and explain its significance in the context of the problem.

Q10. The population growth of a certain species of bacteria can be modeled by the function $f(x) = 2^x$, where x is the time in hours. If $g(x) = x - 3$, find $f(g(x))$ and describe what this function represents in the context of the problem.

Chef's Tips

- Read each question carefully before answering
- Use the given formulas and equations to solve problems
- Check your work and explain your reasoning where necessary